

SQL Code

```
1  USE product_adoption;
2  /* Total Revenue View*/
3  CREATE VIEW vw_total_revenue AS
4  SELECT SUM(final_amount_paid) AS Total_Revenue
5  FROM product_operations;
6
7  /* Total Customers*/
8  CREATE VIEW vw_total_customers AS
9  SELECT COUNT(*) AS Total_Customers
10 FROM customer_profile;
11
12 /* Total Subscriptions*/
13 CREATE VIEW vw_total_subscriptions AS
14 SELECT COUNT(subscription_id) AS Total_Subscriptions
15 FROM product_operations;
16
17 /* Revenue by Plan Category*/
18 CREATE VIEW vw_plan_category_revenue AS
19 SELECT plan_category,
20 COUNT(*) AS Total_Subscriptions,
21 SUM(final_amount_paid) AS Revenue,
22 AVG(final_amount_paid) AS Avg_Revenue
23 FROM product_operations
24 GROUP BY plan_category;
25
26 /* Revenue by Payment Status*/
27 CREATE VIEW vw_payment_status AS
28 SELECT payment_status,
29 COUNT(*) AS Total_Subscriptions,
30 SUM(final_amount_paid) AS Revenue
31 FROM product_operations
32 GROUP BY payment_status;
33
34 /* Customer Segment Revenue */
35 CREATE VIEW vw_customer_segment_revenue AS
36 SELECT c.customer_segment,
37 COUNT(*) AS Total_Subscriptions,
38 SUM(p.final_amount_paid) AS Revenue
39 FROM customer_profile c
40 INNER JOIN product_operations p
41 ON c.customer_id=p.customer_id
42 GROUP BY c.customer_segment;
43
44 SELECT * FROM vw_total_revenue;
45 SELECT * FROM vw_total_customers;
46 SELECT * FROM vw_total_subscriptions;
47 SELECT * FROM vw_plan_category_revenue;
48 SELECT * FROM vw_payment_status;
49 SELECT * FROM vw_customer_segment_revenue;
50
```

```
51  /* window Function*/
52  /* Total Revenue*/
53  SELECT subscription_id, subscription_start_date, final_amount_paid,
54  SUM(final_amount_paid)
55  OVER(
56  ORDER BY subscription_start_date
57  ) AS Running_Revenue
58  FROM product_operations;
59
60  /* Total by Plan Category*/
61  SELECT plan_category, subscription_start_date, final_amount_paid,
62  SUM(final_amount_paid)
63  OVER(
64  PARTITION BY plan_category
65  ORDER BY subscription_start_date
66  ) AS Running_Revenue
67  FROM product_operations;
68
69  /* RANK CUSTOMERS*/
70  SELECT customer_id, final_amount_paid,
71  RANK()
72  OVER(
73  ORDER BY final_amount_paid DESC
74  ) AS Revenue_Rank
75  FROM product_operations;
76
77  /* ROW NUMBER*/
78  SELECT customer_id, subscription_start_date,
79  ROW_NUMBER()
80  OVER(
81  ORDER BY subscription_start_date
82  ) AS Row_No
83  FROM product_operations;
84
85  /* LAG Function*/
86  SELECT subscription_id, subscription_start_date, final_amount_paid,
87  LAG(final_amount_paid)
88  OVER(
89  ORDER BY subscription_start_date
90  ) AS Previous_Revenue
91  FROM product_operations;
92
93  /* LEAD Function*/
94  SELECT subscription_id, subscription_start_date, final_amount_paid,
95  LEAD(final_amount_paid)
96  OVER(
97  ORDER BY subscription_start_date
98  ) AS Next_Revenue
99  FROM product_operations;
100
101  /* Subscription Duration*/
102  SELECT subscription_id, subscription_start_date, subscription_end_date,
103  DATEDIFF(subscription_end_date,subscription_start_date)
104  AS Subscription_Days
```

```
105 FROM product_operations;
106
107 /* Rank By Pla Category */
108 SELECT plan_category, subscription_id, final_amount_paid,
109 RANK()
110 OVER(
111 PARTITION BY plan_category
112 ORDER BY final_amount_paid DESC
113 ) AS Plan_Rank
114 FROM product_operations;
```